

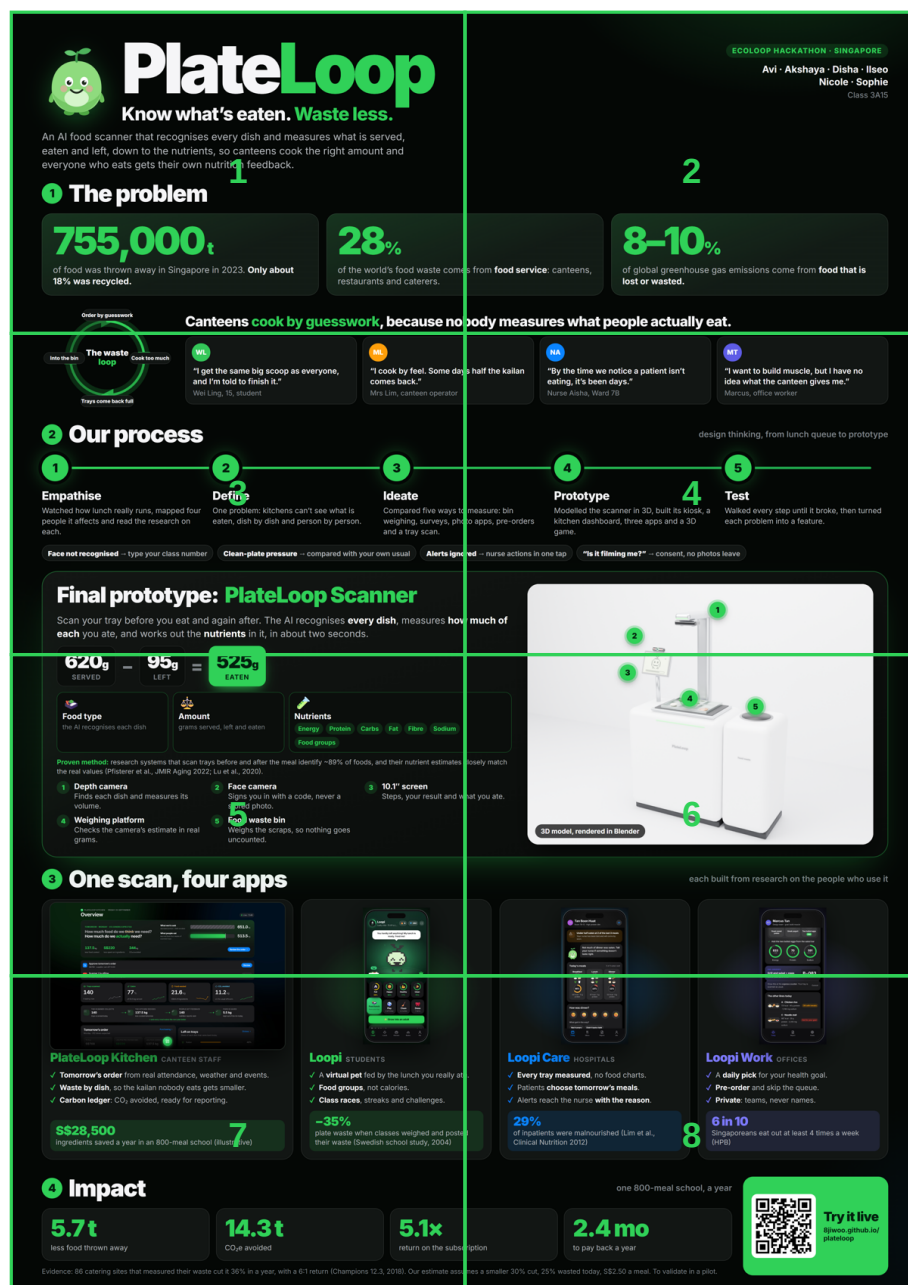
PlateLoop poster: A1 from 8 A4 sheets

Print every sheet on A4 at 100% (Actual size), landscape, with borderless printing on.

Each sheet is exactly one eighth of the A1 poster, edge to edge, so there is nothing to trim.

Lay them out as numbered below (2 across, 4 down) and tape them together from the back.

If your printer cannot print borderless, use the home-printer file instead: it has margins and overlaps.





PlateLoop

Know what's eaten. Waste less.

An AI food scanner that recognises every dish and measures what is served, eaten and left, down to the nutrients, so canteens cook the right amount and everyone who eats gets their own nutrition feedback.

1 The problem

755,000_t

of food was thrown away in Singapore in 2023. **Only about 18% was recycled.**

28%

of the world's food waste comes from restaurants and caterers.

Order by guesswork

Canteens **cook by guesswork**, because not

Avi · Akshaya · Disha · Ilseo
Nicole · Sophie

Class 3A15

from **food service**: canteens,

8–10%

of global greenhouse gas emissions come from **food that is lost or wasted**.

body measures what people actually eat.



WL

"I get the same big scoop as everyone, and I'm told to finish it."

Wei Ling, 15, student

ML

"I cook by feel. Some days come back."

Mrs Lim, canteen operator

2 Our process

1

Empathise

Watched how lunch really runs, mapped four people it affects and read the research on each.

2

Define

One problem: kitchens can't see what is eaten, dish by dish and person by person.

3

Ideate

Compared five ways to weighing, surveys, photos and a tray scan.

Face not recognised → type your class number

Clean-plate pressure → compared with your own usual

Alerts ignored

Final prototype: **PlateLoop Scanner**

Scan your tray before you eat and again after. The AI recognises **every dish**, measures **how much** you ate, and works out the **nutrients** in it, in about two seconds.

s half the kailan

NA

"By the time we notice a patient isn't eating, it's been days."

Nurse Aisha, Ward 7B

MT

"I want to build muscle, but I have no idea what the canteen gives me."

Marcus, office worker

design thinking, from lunch queue to prototype

4

Prototype

measure: bin
to apps, pre-orders

Modelled the scanner in 3D, built its kiosk, a kitchen dashboard, three apps and a 3D game.

5

Test

Walked every step until it broke, then turned each problem into a feature.

red → nurse actions in one tap

"Is it filming me?" → consent, no photos leave

/ much of

2

1



620_g

SERVED

-

95_g

LEFT

=

525_g

EATEN



Food type

the AI recognises each dish



Amount

grams served, left and eaten



Nutrients

Energy

Protein

Carbs

Fat

Fibre

Sodium

Food groups

Proven method: research systems that scan trays before and after the meal identify ~89% of foods, and their nutrient estimates are close to the real values (Pfisterer et al., JMIR Aging 2022; Lu et al., 2020).

1 Depth camera

Finds each dish and measures its volume.

2 Face camera

Signs you in with a code, never a stored photo.

3 10.1" screen

Steps, your result and waste

4 Weighing platform

Checks the camera's estimate in real grams.

5 Food waste bin

Weighs the scraps, so nothing goes uncounted.

3

One scan, four apps

PLATELOOP KITCHEN FRIDAY 25 SEPTEMBER

Overview

Live 11:48

TOMORROW - MONDAY - 812 DINERS EXPECTED

How much food do we ~~think~~ we need?
How much do we **actually** need?

137.5_{kg} **\$5220** **344_{kg}**
less food cooked less spent on ingredients CO₂e avoided

What we'd cook
standard portion x 840 enrolled

What people eat
learned from four weeks of scanned trays

651.0_{kg}

513.5_{kg}

Review the order

Approve tomorrow's order
\$5768 - supplier cut-off 14:00

Review

Scanner 2 is offline

Restart

4 Loopi

Leaky child - 73 XP to L...

4

162

You hardly left anything! My lunch is ready. Feed me!



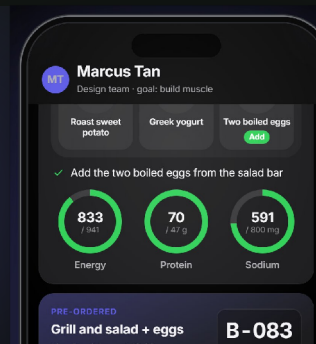
closely match

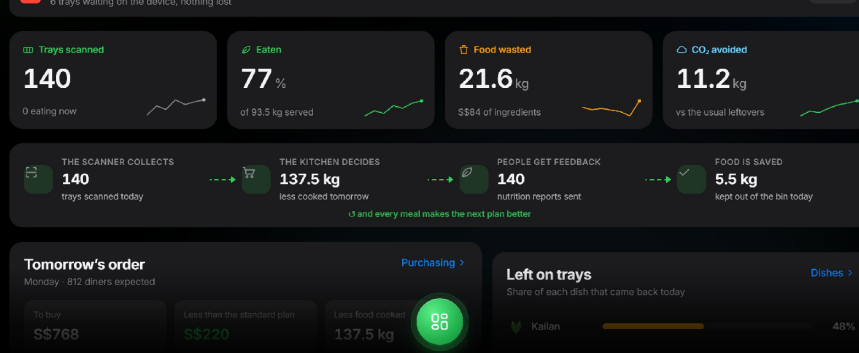
what you ate.



3D model, rendered in Blender

each built from research on the people who use it





PlateLoop Kitchen CANTEEN STAFF

- ✓ **Tomorrow's order** from real attendance, weather and events.
- ✓ **Waste by dish**, so the kailan nobody eats gets smaller.
- ✓ **Carbon ledger:** CO₂ avoided, ready for reporting.

\$528,500

ingredients saved a year in an 800-meal school (illustrative)



Loopi STUDENTS

- ✓ A **virtual pet** fed by the lunch you really ate
- ✓ **Food groups**, not calories.
- ✓ **Class races**, streaks and challenges.

-35%

plate waste when classes weighed and posted their waste (Swedish school study, 2004)

4 Impact

5.7t

less food thrown away

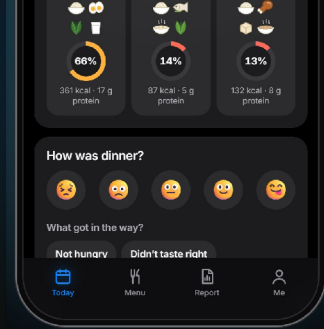
14.3t

CO₂e avoided

5.1x

return on the subscri

Evidence: 86 catering sites that measured their waste cut it 36% in a year, with a 6:1 return (Champions 12.3, 2018). Our estimate assum

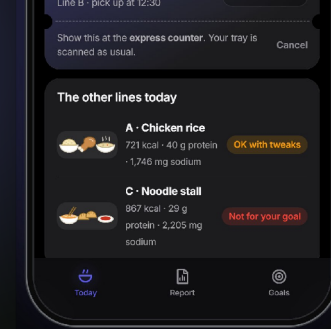


Loopi Care HOSPITALS

- ✓ **Every tray measured**, no food charts.
- ✓ Patients **choose tomorrow's meals**.
- ✓ Alerts reach the nurse **with the reason**.

29%

of inpatients were malnourished (Lim et al., Clinical Nutrition 2012)



Loopi Work OFFICES

- ✓ A **daily pick** for your health goal.
- ✓ **Pre-order** and skip the queue.
- ✓ **Private**: teams, never names.

6 in 10

Singaporeans eat out at least 4 times a week (HPB)

one 800-meal school, a year

2.4 mo

to pay back a year



Try it live

[8jiwoo.github.io/
plateloop](https://8jiwoo.github.io/plateloop)

imes a smaller 30% cut, 25% wasted today, S\$2.50 a meal. To validate in a pilot.